Vehicle routing problem: recent literature review of its variants

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Abstract: The vehicle routing problem is the most studied combinatorial optimisation problem. The purpose of this study is to provide an overview of the research to date in vehicle routing problem variants. The literature is reviewed with a focus on research in three major variants of the vehicle routing problem, namely capacitated vehicle routing problem, mixed depot vehicle routing problem and vehicle routing problem with pickup and delivery. Journal articles from three academic databases, namely Taylor and Francis, Elsevier and Emerald, are selected and reviewed. Ample literature is available on this problem so to restrict the scope, we screened the journal articles using the above mentioned variants precisely, excluding those that are in combination with other variants. This review takes a closer look at 117 research articles selected from various journals. By presenting the past literature, we hope to motivate further research in the field.

Keywords: vehicle routing problems; VRPs; literature review.

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1 Introduction

Supply chain management is the coordination of logistical processes for the transport of products within the supply chain. Logistics management is that part of the supply chain which makes designs, executes and controls the efficient flow and storage of goods, services and related information between the generation point and the point of consumption to meet customer's demands. Supply chain logistics expenditures range from 5% to 50% of a product's total landed cost depending on the industry sector. Vehicle routing problem (VRP) is defined as the task of designing delivery routes to service customers in a company's supply chain. The problem captures the essence of allocation and routing of vehicles at minimal cost, given transportation demand. Hence, it is crucial to effective and efficient logistics management.

The VRP is one of the most challenging problems in the field of combinatorial optimisation. Dantzig and Ramser (1959) first introduced the VRP in 1959. Based on information availability and quality, according to Pillac et al. (2011), there are four categories of VRPs:

- *static and deterministic:* all input is known beforehand and vehicle routes do not change once they are in execution
- *static and stochastic:* input partially known as random variables, realisations of which are only revealed during the execution of the routes
- *dynamic and deterministic:* part or all of the input is unknown and revealed dynamically during the design or execution of the routes
- *dynamic and stochastic:* part or all of the input is unknown and revealed dynamically during the execution of the routes, but in contrast with the previous category, exploitable stochastic knowledge is available on the dynamically revealed information.

The VRP is defined as a type of combinatorial problem. Hence, heuristics and meta-heuristics are used to get realistic and close to the optimal solution. Researchers presently are making use of combinations of different heuristics to shorten the time needed for finding the nearly optimal solutions. Recent literature concerning the above mentioned problem includes Geetha et al. (2014) in which both k-means algorithm and nested particle swarm algorithm are used for solving multi-depot VRP, Avci and Topaloglu (2015) used simulated annealing with variable neighbourhood descent, similarly for simultaneous pickup and delivery vehicle routing variant, hybrid heuristic, based on iterated local search (ILS) and GENIUS is used by Marcone et al. (2012).

This article is organised as follows. First, model formulations for the most general form of VRP are presented in order to clearly define the problem. Then, for each problem variant, a brief description is given. Then, the methodology of review is explained. In the next section, detailed exploration of research proposed in the literature concerning the pre-defined variants is done. This is followed by description of benchmark algorithms used. Finally, the findings are discussed and conclusion section gives directions for future research.

2 Vehicle routing problem

2.1 Mathematical formulation {most general form}

To obtain a mathematical formulation of the basic VRP, first some notation and definitions are needed. This paper uses the same formulation as the one found in Baldacci et al. (2008). It is the most general form of VRP stated as heterogeneous VRP with fixed costs and vehicle dependent routing costs. The basic VRP is given by a set of identical vehicles, a depot, a set of customers to be visited and a directed network connecting the depot and customers.

Let $G = (V, A)$ be a directed graph with vertex set $V = \{0,\ldots\ldots,n\}$ and arc set A.

- Vertex 0 corresponds to the depot.
- Vertices 1,2,……,n correspond to the customers, take $V' = \{1,\ldots\ldots,n\}$.
- A set of $M = \{1,\ldots,m\}$ identical vehicles, each vehicle with capacity C, is available at the depot.
- The vehicles must return to the depot they originated from.
- Each customer $i \in V'$ is associated with a known demand $d_i \geq 0$ to be delivered (assume $d_0 = 0$ and $d_i \leq C$ for all $i \in V'$).
- A fixed cost F for each vehicle required that models the rental or capital amortisation cost of a vehicle.
- A non-negative cost c_{ij} is associated with each arc $(i, j) \in A$ representing the travel cost between vertices i and j (assume $c_{ii} = 0$).

- Only one vehicle handles the deliveries for a given customer.
- The number of vehicles is equal to the number of routes, meaning that a vehicle can only complete one route per day.
- The travelling distance between customers *i* and *j* is the same as the travel distance between *j* and *i*.
- The demand of the customers on every route is known with certainty. The demand of the customers in total on one route cannot exceed the capacity of the specific vehicle that will cover that route.

There are two types of decision variables in a VRP:

- the binary variables x_{ij} are used as vehicle flow variables that take value 1 if a vehicle travels directly from customer *i* to customer *j*, and 0 otherwise
- the flow variables y_{ij} specify the quantity of goods that a vehicle carries when it leaves customer *i* to service customer *j*.

With this notation, the mathematical formulation of the VRP becomes

minimise:

$$\sum_{j\in V'} Fx_{0j} + \sum_{i,j\in V, i\neq j} c_{ij}x_{ij}, \tag{1}$$

subject to:

$$\sum_{i\in V} x_{ij} = 1 \qquad \forall j \in V' \tag{2}$$

$$\sum_{i\in V} x_{ip} - \sum_{j\in V} x_{pj} = 0 \qquad \forall p \in V' \tag{3}$$

$$\sum_{j\in V'} x_{0j} \leq m \tag{4}$$

$$\sum_{i\in V} y_{ij} - \sum_{i\in V} y_{ji} = d_j \qquad \forall j \in V' \tag{5}$$

$$d_j x_{ij} \leq y_{ij} \leq (C - d_i)x_{ij} \qquad \forall i,\ j \in V, i \neq j \tag{6}$$

$$y_{ij} \geq 0 \qquad \forall i,\ j \in V, i \neq j \tag{7}$$

$$x_{ij} \in \{0, 1\} \qquad \forall i,\ j \in V, i \neq j \tag{8}$$

In the above formulation, the objective function (1) is to find a collection of simple circuits in the graph *G* (each circuit corresponding to a vehicle route) with minimal cost (vehicle cost and travel cost) such that:

- each customer is visited once by exactly one vehicle route {constraint (2)}
- if a vehicle visits a customer, it must also depart from it {constraint (3)}

- the number of vehicle routes is not greater than the number of available vehicles, i.e., there is a finite vehicle fleet {constraint (4)}
- the difference between the quantity of goods a vehicle carries before and after visiting a customer is equal to the demand of that customer {constraint (5)}
- the sum of the demands of the customers visited by each vehicle route does not exceed the given vehicle capacity C {constraint (6)}
- the decision variable y_{ij} is positive{constraint (7)}
- the decision variables x_{ij} is binary{constraint (8)}

2.2 Variants of VRP

2.2.1 Capacitated vehicle routing problem

The problem in which fixed vehicles of uniform capacity is used to service known customer demands from a common depot at minimum cost is termed as capacitated vehicle routing problem (CVRP). Basically we have a depot, a set of customers, a set of vehicles as in the case of classical VRP but the difference is that in the CVRP each vehicle has a fixed capacity. One more variant is cumulative CVRP in which the sum of arrival times at customers is minimised, instead of the classical route length.

2.2.2 Multiple depot vehicle routing problem

In these types of problems, vehicles provide services to customers from several depots. It involves the assignment of customers to depots. A fleet of vehicles is based at each depot. Each vehicle after servicing the customers assigned to that depot, return to the same depot. It can be solved in two stages: first, customers must be allocated to depots then routes must be constructed that link customers which have the same depot allocated to them.

2.2.3 Periodic VRP

Here the planning period is extended to number of days instead of a single day in case of original VRP. It is not necessary that the customers are visited each day. Customers can have different delivery frequencies.

2.2.4 Split delivery VRP

If the size of customer orders exceeds the size of vehicle, then the customer needs to be served by different vehicles. The objective is to reduce the cost, number of vehicles and total travel time.

2.2.5 VRP with satellite facilities

The VRP with satellite facilities is an important variant of the VRP in which a commodity is provided to the customers consistently via a central facility.

Two-echelon VRP

Two-echelon VRP considers two levels. The first level is the delivery from the central depot to the satellite facilities and the second level is the delivery from the satellites to the customers. A limit on the number of vehicles is imposed at each level. The objective is to minimise the total routing cost of the system.

2.2.6 Heterogeneous fleet VRP

This variant of the classical VRP allows vehicles with different capacities or different fixed cost, which is not the case in homogeneous vehicle fleet.

2.2.7 VRP with time windows

In this type of VRP, time window is linked with each customer, defining an interval in which customer has to be supplied with the supplies. Each route must start and end within the time window associated with the depot. The real life application of this includes milk or perishable product's delivery system etc. The VRP with time windows is an extension to the CVRP, where time dimension constraints have been incorporated. A recent survey on solving large-scale VRPTWs is carried out by Gendreau and Tarantilis (2010).

2.2.8 Green VRP

This variant of the VRP aims to include different environmental issues in the optimisation procedure. They have the objective to balance environmental and economic costs. Detailed survey on green VRP can be found in Demir et al. (2014). Some of the important work includes greenhouse gas emissions and the effects of using 'greener' fleet configurations (Juan et al., 2014) etc.

2.2.9 Generalised VRP

The generalised VRP is an extension of the VRP in which set of vehicle routes through a set of clusters have to be constructed such that the tour includes exactly one vertex from each cluster and satisfies capacity constraints. Here the customers are partitioned into clusters and vehicles are obligated to visit only one customer in each cluster.

2.2.10 Stochastic VRP

When one or several constituents of the problem are random, the problem is known as stochastic VRP. Either the availability of customer, product demand or travel time/service time can be a random variable. Two phase solution in which first a solution is determined without considering random variables and then a corrective action can be taken when the values of the random variables are known. Here also the objective is to minimise the vehicle fleet and the sum of travel time needed to supply all customers with random values on each execution for the customers to be served, their demands and/or the service and travel times.

2.2.11 Rich VRP

Rich vehicle routing problem (RVRP) was first defined by Toth and Vigo (2001) that include most of the mentioned constraints and situations and also other real-world constraints and situations.

2.2.12 Open VRP

In this type of problem, vehicles do not need to return to the depot after completing their delivery services. The industries which do not have their own fleet of vehicles, third party logistics providers are used there. In these cases, vehicles need not return to the central depot after their deliveries. Each route in the open VRP is a Hamiltonian path instead of a cycle.

Boden et al. (1983) presented a paper in which FedEx delivery routes are studied. Another study of FedEx home delivery service to customers and newspaper home delivery problem was done by Levy et al. (2005). These are some of the real life applications of open VRP.

2.2.13 VRP with pick-up and delivery

The VRP with pickup and delivery is a variant of the VRP where the vehicles are not only required to deliver goods to customers but also to pick some goods up from the customer locations (Subramanian et al., 2010).

VRP with simultaneous pickup and delivery

This type of VRP constitutes the customer that will have both a delivery as well as a pickup demand. So here we need to take into consideration that goods that customers return to the delivery vehicle must fit into it.

VRP with backhauls

This is a VRP in which customers can demand or return some products. But here we assume that all deliveries must be made on each route before any pickups can be made. In this case, the goods that customers return to the delivery vehicle must fit into it.

VRP with mixed pickup and delivery

It is the special case of VRP with pickup and delivery in which customers either have a pickup or a delivery demand and the pickups and deliveries can occur in any sequence.

2.2.14 Asymmetric VRP

The cost for going from customer i to j is different from that for going from j to i due to certain road constraints such as one-way directions or the presence of tolls etc.

3 Scope and methodology of the survey

Reviewed literature is confined as follows:

The objective of this study is to compile the available literature on CVRP, multiple depot vehicle routing problem (MDVRP) and VRP with pickup and delivery. We used the following databases for selecting the relevant papers, Taylor and Francis, Elsevier and Emerald, keeping 'CVRP', 'MDVRP' and 'VRP with pickup and delivery' as title words, and the source type to be English language journal articles. Also some of the research articles were taken from Google Scholar according to their relevancy and impact on the pre-defined subject. Bibliographies of survey papers were also taken into consideration for some relevant articles. Since the VRP already is an extensive problem, the decision was made not to include any related problems, such as combination of other variants of VRP with the three variants that are studied exclusively. The literature we reviewed is spread over a large period of time i.e. from 1959 to 2015, but we gave more importance to recent literature from 2011 onwards.

To extract only the actual relevant literature, all available journals were ranked according to their impact factor. From the suggested journals, some journals were selected because of their importance in the domain of either Operational Research or Logistical Management. After the selection process, analysis of abstracts was done in order to ensure that the central theme was relevant with the concerned subject. If the scope was not clear by the abstract, then the article was analysed in detail to gauge the significance of the study.

Following this methodology, a total of 117 research articles were screened, 37 are from CVRP, 19 are from MDVRP and 61 from VRP with pickup and delivery.

4 Literature review

In the field of combinatorial optimisation, vehicle routing and scheduling problems have been investigated widely. Different variants of the basic VRSP are examined that involve the consideration of additional constraints or other variations in the configuration of the appropriate model. We are reviewing the literature on three variants of VRP. For extensive review of the variants of VRP, Subramanian (2012) can be referred.

4.1 Capacitated vehicle routing problem

The problem in which a fixed fleet of supply vehicles of uniform capacity must service known customer demands for a single product from a common depot at minimum cost. In the CVRP we are given a depot, a set of n customers, a set of m vehicles and a distance measure as in the m-TSP, but in the CVRP every vehicle has a capacity Q and every customer $i \in \{1,\ldots,n\}$ has a demand q_i.

In the homogeneous CVRP, each vehicle has the same capacity Q. The only difference is that if the total demand of all customers on a route does not exceed the vehicle capacity, then only the route is considered feasible i.e. $(\sum_{j=1}^{n} d_j \leq Q)$ (where d_j is the demand of customer v_j). Also the total capacity of vehicles should be greater than the total demands of all customers on that route. While in the heterogeneous CVRP, the vehicles have their own capacity each.

Research till 2011

A lot of exact algorithms had been suggested in the past literature to solve the CVRP. The algorithm used in these are branch and cut which combines the branch and bound and cutting plane method, developed by Lysgaard et al. (2004), branch and cut based on the two-commodity flow formulation proposed by Baldacci et al. (2004) and branch cut and price which is a combination of branch and cut and column generation method proposed by Fukasawa et al. (2006), algorithm based on set partitioning with extra cuts by Baldacci et al. (2008). After that Pessoa et al. (2008) extended the branch cut and price algorithm to propose a new one with new families of cuts which also deals with worst complexities of pricing problem. Ralphs et al. (2003) implemented a branch and cut algorithm based on the two-index formulation. A recent work of Contardo et al. (2014) reports an exact algorithm based on cut-and-column generation. Past surveys of exact algorithms for CVRP was done by Chicago Laporte and Nobert (1987) and Baldacci et al. (2007, 2010).

We now discuss some heuristics algorithms proposed to solve the CVRP.

The algorithms used are:

Taillard (1993) used tabu search algorithm. After that Rochat and Taillard (1995) proposed a method based on the combination of tabu search and adaptive memory procedure algorithm. Tarantilis (2005) also applied similar methodology called solutions elite parts search (SEPAS). Derigs and Kaiser (2007) used a version of tabu search algorithm on CVRP which is called attribute-based hill climber heuristic originally proposed by Whittley and Smith (2004) who described it as parameter-free local search heuristic. Reimann et al. (2004) proposed an ant colony optimisation heuristics which produces optimum solutions by mimicking the nature of ants. Prins (2004) proposed a memetic algorithm which is similar to genetic algorithm but in MA memes are used to transfer information instead of genes. Nagata (2007) also used the same method with refurbishing involved in it and Nagata and Bräysy (2009) extended it in which local searches were restricted using some techniques.

De Franceschi et al. (2006) used integer linear programming (ILP) heuristic. An extension of large neighbourhood search was proposed by Pisinger and Røpke (2007). This algorithm selects from a number of insertion and removal heuristics, to make the search stronger. Kytöjoki et al. (2007) developed an algorithm comprising of variable neighbourhood search (VNS) and guided local search metaheuristic. Mester and Bräysy (2007) presented a two-phase heuristic which combines the strengths of guided local search, large neighbourhood search and evolutionary strategies. Toth and Tramontani (2008) presented an algorithm based on the exploration of an exponential neighbourhood by solving an ILP problem. Prins (2009) proposed some heuristics based on ILS, greedy randomised adaptive search procedure (GRASP) and evolutionary local search.

Chen et al. (2010) suggested a hybrid method known as iterated variable neighbourhood descent in which to escape from local minima, cross exchange technique is used. Zachariadis and Kiranoudis (2010) developed an adaptation based on tabu search, guided local search and VNS. Vidal et al. (2012) came up with a new hybrid genetic search algorithm with adaptive diversity control. A detailed outline of the classical heuristics was developed by Laporte and Semet (2002), while the review of use of metaheuristics to the CVRP was done by Cordeau et al. (2005) and Gendreau et al. (2002).

A compilation of relevant research papers after 2011 can be found in Table 1.

Table 1 Capacitated vehicle routing problem

S. no.	*Algorithm*	*Author*	*Description*
1	Artificial bee colony	Szeto et al. (2011)	An artificial bee colony heuristic is based on the behaviour of honey bees. It consists of three steps: transfer of employed bees to the food to determine their nectar quantity; selection of the food sources-based on the information given by the employed bees; determining the pathfinder bees to look out for possible food sources.
2	Branch-and-cut + adaptive memory programming metaheuristic	Gounaris et al. (2013)	A combination of branch-and-cut method based on two-commodity flow representation and new heuristics to separate cuts and adaptive memory programming metaheuristic algorithm to detect fairly good solutions is used.
3	Genetic algorithm	Nazif and Lee (2012)	In this algorithm, the objective is to minimise cost. To do that, proposed algorithm uses a modified crossover operator designed by a complete undirected two-part graph.
4	Nested particle swarm optimisation	Kanthavel and Prasad (2011)	The algorithm is used as master particle swarm optimisation and slave particle swarm optimisation for the picking out candidate list and route succession to optimise the mode.
5	Multiple phase neighbourhood search + GRASP	Marinakis (2012)	In this method, Lagrangean relaxation and subgradient optimisation is utilised for selecting terminating conditions. Also a new strategy, the circle restricted local search moves strategy, is used to further explore the neighbourhood.
6	Parallel tabu search algorithm	Jin et al. (2012)	In this algorithm, mixed use of single and combined neighbourhood which are embedded in the tabu list is done to provide solutions.
7	Granular tabu search	Escobar (2013)	In the hybrid heuristic algorithm proposed here, initial solution is constructed after which a modified granular tabu search, is applied to improve it. Also, a random perturbation method is used to make the jump out of the local optimum.
8	Particle swarm optimisation	Kim and Son (2012)	Here particle encoding and decoding is done by probability matrix.
9	Multiple ant colony optimisation algorithm	Ting and Chen (2013)	Two phase method in which initially the given problem is divided into facility location problem and MDVRP. Then, the algorithm is used for location selection, customer assignment, and VRP.
10	Cooperative parallel metaheuristic	Jin et al. (2014)	Multiple parallel tabu search threads that cooperate by asynchronously exchanging best-found solutions through a common solution pool. The solutions sent to the pool are clustered according to their similarities.
11	Improved branch-cut-and-price	Pecin et al. (2014)	The author introduces a technique for greatly decreasing the impact on the pricing of the subset row cuts, thus allowing much more such cuts to be added. Also it contains route enumeration and strong branching techniques.

Table 1 Capacitated vehicle routing problem (continued)

S. no.	*Algorithm*	*Author*	*Description*
12	Multiple neighbourhood search	Barbucha (2014)	To avoid the local minima and to get the fairly optimum solutions, multiple neighbourhoods are explored. So all the agents searching in multiple parts of the solution space cooperate and share information with each other to solve the problem
13	VNS algorithm	Yu and Maghfiroh (2014)	VNS with path-relinking is used, to combine intensification and diversification strategies.
14	GRASP + ILP hybrid algorithm	Contardo et al. (2013)	Three-phase heuristic, GRASP followed by local search procedures to construct a bundle of solutions; an integer-linear program with the objective of constructing a new solution as a combination of the routes belonging to the solutions of the bundle taken as input; the same ILP is iteratively solved by column generation to improve the solutions found during the first two stages.

4.2 Multiple depot vehicle routing problem

In multiple depot vehicle routing problems, customers may be serviced by any of the depots. So, this distribution problem is a set of independent VRPs. It involves the assignment of customers to depots. A fleet of vehicles is based at each depot. Each vehicle after servicing the customers assigned to that depot, return to the same depot. It can be solved in two stages: first, customers must be assigned to depots then routes must be made that link customers assigned to the same depot.

Research till 2011

Tillman (1969) proposed an algorithm based on savings heuristic to solve MDVRP. After that Tillman and Hering (1971) improved the original algorithm with look ahead selection procedure. Tillman and Cain (1972) also added up to the original algorithm using an upper bound algorithm. Gillet and Johnson (1976) introduced the multi-terminal sweep algorithm based on classical sweep algorithm of Gillet and Miller. Raft (1982) recommended a modular algorithm which initially separates the problem, solve it part by part and finally combine it to solve the MDVRP. Chao et al. (1993) developed an algorithm that improves the solution generated using the savings heuristic of Golden et al. Renaud et al. (1996a, 1996b) employed a TS heuristic. Cordeau et al. (1997) suggested a tabu search algorithm that uses a generalised insertion procedure (GENI) mechanism. Pisinger and Ropke (2007) proposed the adaptive large neighbourhood search. Baldacci and Mingozzi (2009) proposed the exact algorithm that uses bounding procedures based on the LP-relaxation and Lagrangean relaxation. Vidal et al. (2012) came up with a new hybrid genetic search algorithm with adaptive diversity control.

In Table 2, research papers are summarised after 2011.

Table 2 Multiple depot vehicle routing problem

S. no.	*Algorithm*	*Author*	*Description*
1	Parallel improved ant colony optimisation	Yu et al. (2011)	An improved ant colony optimisation with coarse-grain parallel strategy, ant-weight strategy and mutation operation, is used to compare 23 benchmark problems with those of other ant colony optimisations.
2	Ant colony optimisation	Xie et al. (2012)	Graphs were made from customers assigned to vehicles and ACO was applied to the graph to optimise the MDVRP. Finally, reallocating of customers was done using TSP.
3	ILS meta-heuristic with biased-randomised processes	Juan et al. (2012)	ILS consists of two biased randomised approaches to define the operator defined to escape local minima and to generate optimal solutions. The biased-randomisation processes rely on the use of a pseudo-geometric probability distribution.
4	Shuffled frog leaping algorithm	Luo and Chen (2014)	Power law extremal optimisation neighbourhood search is introduced to further enhance the search ability of shuffled frog leaping algorithm that simulates the memetic evolution of a group of frogs.
5	Hybrid genetic algorithm + adaptive population diversity management procedures	Vidal et al. (2012)	The algorithm combines the search criteria of population-based evolutionary search, the improvement abilities of neighbourhood-based metaheuristics, and modified population-diversity management schemes.
6	Bi-level Voronoi diagram-based metaheuristic	Tu et al. (2013)	As the name suggests, two levels are used. Customers are allocated to depots in the first level. The lower level restricts the search criteria of reallocating customers among the depots and reorganise the customers from each depot to its neighbours.
7	Nested particle swarm optimisation	Geetha et al. (2013)	MDVRP is divided into multiple VRP by improved k-means clustering algorithm. Then each VRP is solved by nested particle swarm optimisation with genetic operators.
8	Hybrid granular tabu search algorithm	Escobar et al. (2014)	Granular tabu search procedure is used for improving the solutions obtained by a hybrid procedure.

4.3 VRP with pick-up and delivery

The VRP with pickup and delivery is a variant of the VRP where the vehicles are required not only to deliver goods to customers but also to pick some goods up from the customer locations. This imposes both a coupling constraint and a precedence constraint on the original VRP. The coupling constraint is that each pickup and delivery location of a single request must be visited exactly once by the same vehicle. The precedence constraint is that the location of the request's pickup must be serviced before its corresponding delivery. A classification survey by Berbeglia et al. (2007) provides

different variants of VRP with pickups and deliveries. Some important ones are discussed here:

4.3.1 VRP with simultaneous pickup and delivery

This type of VRP constitutes the customer that will have both a delivery as well as a pickup demand. So here we need to take into consideration that goods that customers return to the delivery vehicle must fit into it.

Min (1989) was the first to introduce three phase heuristics for VRP with simultaneous pickup and delivery. His study considered distribution problem of a public library with 22 branch libraries and two vehicles based on the central library. Halse (1992) used cluster first route second algorithm for the same. Salhi and Nagy (1999) used insertion-based heuristics and Dethloff (2001) used constructive heuristics. Angelelli and Mansini (2002) proposed branch cut and price for the special case of simultaneous pickup and delivery problem with time window constraint. Vural (2003) exercised genetic algorithm and Ropke and Pisinger (2006) used large neighbourhood search. Crispim and Brandão (2005) implemented a combination of tabu search and variable neighbourhood descent algorithm. Most of the algorithms used for this type of variant are hybrid in nature. Chen and Wu (2006) work consists of the combination of insertion-based procedure, record to record travel, tabu search and route improvement procedures. Chen et al. (2006) again contributed by making use of tabu search and simulated annealing. Montané and Galvao (2006) made use of tabu search algorithm. An exact branch and price algorithm was proposed by Dell'Amico et al. (2006). Bianchessi and Righini (2007) analysed a new hybrid algorithm involving constructive search, tabu search and VNS heuristic, Zachariadis et al. (2009) involving tabu search and guided local search, Gajpal and Abad (2009) using ant colony optimisation, Zachariadis et al. (2010) with tabu search and adaptive memory procedure method are some of the heuristics proposed for this variant.

Table 3 VRP with simultaneous pick-up and delivery

S. no.	*Algorithm*	*Author*	*Description*
1	Branch-and-cut algorithm	Subramanian et al. (2011)	Branch-and-cut approach was proposed over a formulation with only the edge variables and additional route feasibility constraints are also implemented in a lazy fashion.
2		Karaoglan et al. (2011)	Branch-and-cut algorithm and a local search based on simulated annealing algorithm to obtain upper bounds is applied to solve VRP.
3	Local search metaheuristic	Zachariadis and Kiranoudis (2011)	The proposed algorithm explores solution neighbourhoods and to avoid getting in earlier solution neighbourhood, it makes use of a model based on aspiration criteria of tabu search.
4	Genetic algorithm-based heuristic	Tasan and Gen (2012)	A genetic algorithm-based approach that use the natural principle of a genetic population, comprising of an initial population generator, appropriate fitness function, and genetic operators such as crossover, mutation and inversion is used in this study.

Table 3 VRP with simultaneous pick-up and delivery (continued)

S. no.	*Algorithm*	*Author*	*Description*
5	ILS + GENIUS + variable neighbourhood descent	Souza et al. (2011)	Hybrid algorithm, called GENILS, using the combination of ILS, variable neighbourhood descent and GENIUS is used to improve the results of test sets.
6	Branch-cut-and-price	Subramanian et al. (2012)	Branch-cut-and-price approach composed of column generation and cutting planes is used to find the lower bound.
7	Particle swarm optimisation + variable neighbourhood descent algorithm	Goksal et al. (2013)	Particle swarm optimisation is used in which a local search is performed by variable neighbourhood descent algorithm. It also implements an annealing-like approach to maintain the level of swarm diversity.
8	Artificial immune systems-based algorithm	Yu and Lau (2013)	To generate initial routes, sweep method based on polar angles is used. For the mutation operator, a variety of local search techniques i.e. six inter-route local search methods and four intra-route local search methods are applied to understand the diversity.
9	Multi-start simulated annealing heuristic	Yu and Lin (2014)	Algorithm for solving the problem comprises of two stages: multi start simulated annealing algorithm to generate solutions and then hill climbing strategy to find the local minima.
10	Modified tabu search and ant system	Yousefikhoshbakht et al. (2014)	To get both the global-best and iteration-best solutions, the size of the tabu list is considered as a variable for intensification and diversification policies. Also to improve the best found solution in each iteration, the ant algorithm is used for each route of the solution.
11	Adaptive local search algorithm	Avci and Topaloglu (2015)	Two phase algorithm. Local search solution approach is developed which is a combination of a simulated annealing with variable neighbourhood descent. To further improve the solution, VND is used with perturbation techniques.

4.3.2 VRP with backhauls

This is a VRP in which customers can demand or return some products. But here we assume that all deliveries must be made on each route before any pickups can be made. In this case, the goods that customers return to the delivery vehicle must fit into it. This is due to the fact that the vehicles are rear-loaded, and reshuffle of the loads on the tracks at the delivery points is not considered economical or feasible. Also, the quantities to be delivered and picked-up are fixed and known in advance.

Deif and Bodin (1984) extended savings algorithm to solve VRP with backhauls in which they used penalty terms to push backhauls to the end of routes. Yano et al. (1987) used an exact branch and bound algorithm based on set covering. Goetschalckx and Jacobs-Blecha (1989) used route first cluster second algorithm based on space-filling curves and then the solution is improved. Min et al. (1992) developed a decomposition

approach involving grouping, allocating and sequencing. Anily (1996) proposed a two stage clustering and allotting algorithm. Toth and Vigo (1996, 1997) proposed first a cluster-first-route-second heuristic, and then a branch-and-bound exact approach that uses a Lagrangian lower bound. Gendreau et al. (1997) used minimum spanning tree to transform the problem. Osman and Wassan (2002) developed a method based on savings and reactive tabu search. Liu et al. (2010) also used tabu search algorithm to minimise the used vehicle number and total travel distance.

The research papers are discussed in Table 4:

Table 4 VRP with backhauls

S. no.	*Algorithm*	*Author*	*Description*
1	Local search metaheuristic	Zachariadis and Kiranoudis (2012)	In this algorithm, solution neighbourhood which are composed of chain of customers of different lengths are explored by using Fibonacci heaps-based search.
2	Integer programming heuristic	Yalcin and Erginel (2012)	A heuristic based on integer programming is proposed. Firstly delivery and backhaul customers are assigned to vehicles with the integer model. Then routes are made and finally local search is used to improve them.
3	Multi-objective evolutionary approach	Garcia-Najera and Bullinaria (2012)	Problems with multi objective function are solved with an evolutionary algorithm.
4		García-Nájera et al. (2015)	In this paper, a similarity-based selection evolutionary algorithm approach is proposed for finding improved multi-objective solutions.
5	ILS algorithm	Cuervo et al. (2014)	Local search heuristic is used which explores a broad neighbourhood structure at each iteration and stores information and keeps on shuffling between feasible and infeasible subset of solutions.
6	Fuzzy multi-objective programming algorithm	DincYalcın and Erginel (2015)	A new algorithm is proposed, which has three phases – clustering, routing and local search. The objective here is to minimise the total distance and maximise the total savings value.

4.3.3 VRP with mixed pickup and delivery

It is a special case of VRP with pickup and delivery in which customers either have a pickup or a delivery demand and the pickups and deliveries can occur in any sequence. Basically, it can be referred as a special case of VRP with simultaneous pickup and delivery in which only one of the demand can be serviced.

The first work related to this was proposed by Golden et al. (1985) called backhaul insertion procedure in which customers for pickups i.e. backhaul customers were inserted in the routes formed for deliveries to customers. After that Casco et al. (1988) presented the load-based backhaul insertion heuristic. The other heuristics are: Mosheiov (1998) developed a method in which tour is partitioned into sub tours which are served by vehicles and called it two tour partitioning heuristics, Salhi and Nagy (1999) used cluster insertion heuristics in which more than one pickups are inserted at a time, Dethloff (2002) proposed insertion heuristic based on residual capacities, Wade and Salhi (2004) used ant colony optimisation which simulate the nature of ants to find the optimal routes

with modified trail updating methods, Ropke and Pisinger (2006) applied large neighbourhood search technique which works by removal of vertices and then insertion of vertices into the route and then it is evaluated, Nagy and Salhi (2005) developed a construction with improvement heuristic algorithm. Crispim and Brandão (2005) introduced a hybrid algorithm comprised of tabu search and variable neighbourhood descent algorithm, Wassan et al. (2008a, 2008b) work based on reactive tabu search, Gajpal and Abad (2009) again implemented ant colony optimisation in which local search is based on customer insertion and sub path exchange multi route method. Tütüncü et al. (2009) implemented a visual interactive approach that is based on greedy randomised adaptive memory programming search (GRAMPS).

The research papers after 2011 are given in Table 5:

Table 5 VRP with mixed pickup and delivery

S. no.	*Algorithm*	*Author*	*Description*
1	Memetic algorithm	Karaoglan and Altiparmak (2015)	Memetic algorithm which combines local search with crossover, is implemented to solve the problem and results are compared with lower bounds obtained by Branch and cut algorithm.
2	Adaptive local search algorithm	Avci and Topaloglu (2015)	Two-phase algorithm. Local search solution approach is developed which is a combination of a simulated annealing with variable neighbourhood descent. To further improve the solution, VND is used with perturbation techniques.

5 Solution methods

There are a lot of algorithms to solve variants of VRP. Most of them were reported in Cordeau et al. (2006) and were described here as well.

5.1 Exact approaches

A large number of exact algorithms have been proposed for the VRP. These types of approaches propose to compute every possible solution until the bests is obtained.

5.1.1 Branch and bound

A branch and bound algorithm uses a divide and conquer strategy to partition the solution space into sub problems and then optimises individually over each sub problem. Using branch and bound, we initially examine the entire solution space. In the processing phase, we relax the problem. In doing so, we admit solutions that are not in the feasible set. A lower bound of an optimal solution is determined by solving the relaxation. If the solution comes out to be an element of feasible set or has cost equal to that of feasible set cost, then we are done the new solution is optimal.

5.1.2 Branch and cut

Branch and cut are exact algorithms consisting of a combination of a cutting plane method with a branch-and-bound algorithm. The method solves the linear program without the integer constraint using the regular simplex algorithm. When an optimal solution is obtained, and this solution has a non-integer value for a variable that is supposed to be integer, a cutting plane algorithm may be used to find further linear constraints which are satisfied by all feasible integer points but violated by the current fractional solution.

5.1.3 Branch cut and price

A new exact algorithm for the VRP has been presented by Fukasawa et al. (2006). This method has been judged as the best exact method (Baldacci et al., 2007). The main idea is to combine the branch-and-cut approach with the *q*-routes approach to derive superior lower bounds. Thus, the branch and cut and price algorithm. The idea of combining column and cut generation to improve lower bounds has rarely been used, since new dual variables corresponding to separated cuts may have the undesirable effect of changing the structure of the pricing sub-problem. However, if cuts are expressed in terms of variables from a suitable original formulation, they can be incorporated into the column generation process without disturbing the pricing.

5.2 Heuristics

Heuristic methods perform a relatively limited exploration of the search space and typically produce good quality solutions within modest computing times.

5.2.1 Constructive methods

Gradually build a feasible solution while keeping an eye on solution cost, tour construction algorithms never try to improve the solution, they stop as soon as it is found. Many construction heuristics for VRPs fall into one of the three classes: insertion heuristics, savings heuristics and clustering heuristics.

5.2.1.1 Savings: Clark and Wright

This classical algorithm was first proposed in 1964 by Clarke and Wright to solve CVRPs. It expresses the cost savings obtained by joining two routes into one route. The method starts with vehicle routes containing the depot and one other vertex. All vertices are visited by separate route from depot.

According to Labbé et al. (1991),

1 Compute the savings $s_{ij} = c_{i0} + c_{0j} - c_{ij}$ for $i, j = [1,\ldots,n]$ where 0 is the depot and i, j are two vertices. Create $n - 1$ vehicle routes $[0, i, 0]$ $[i = 1,\ldots,n]$.

2 Order the savings in a non-increasing fashion.

3 Consider two vehicle routes containing arcs $[i, 0]$ and $[0, j]$, respectively. If $s_{ij} > 0$, tentatively merge these routes by introducing arc $[i, j]$ and by deleting arcs $[i, 0]$ and

[0, j]. Implement the merge if the resulting route is feasible. Repeat this step until no further improvement is possible. Stop.

5.2.1.2 Two-phase algorithm

The problem is decomposed into its two natural components: clustering of vertices into feasible routes and actual route construction, with possible feedback loops between the two stages.

Cluster-first, route-second algorithms

In these type of algorithms, as the name suggests, clusters are formed first by grouping customers and then the routes are determined by suitably sequencing the customers within each cluster.

5.2.1.2.1 The sweep algorithm

The sweep algorithm is proposed by Wren (1971) and Wren and Holliday (1972) for CVRPs with one or several depots, and vertices located in the Euclidean plane. Gillett and Miller (1974) gave the algorithm its name.

In this algorithm, polar coordinates of the vertices are used. The algorithm starts with an arbitrary vertex. The remaining vertices are ranked in the increasing order of polar angle. Then the remaining vertices are assigned to the current vehicle by considering them in order of increasing polar angle with respect to the depot and the initial customer as long as the capacity is not surpassed, after which a new route is initialised. Once all customers are assigned to vehicles, each route is separately defined by solving a TSP.

5.2.1.2.2 Fisher and Jaikumar

Fisher and Jaikumar (1981) proposed generalised assignment problem (GAP) to form clusters and then solve a TSP for each cluster. In GAP minimum cost assignment of items to a given set of bins of capacity Q is done where weight is the parameter to characterise items and an assignment cost for each bin. A customer, termed as a seed, is assigned to each vehicle and the assignment cost of a customer to a vehicle is calculated which is equal to its distance to the seed. The given problem is then either optimally or heuristically solved and a TSP on each cluster is solved to determine the final routes.

5.2.1.2.3 The petal algorithm

A different family of two-phase methods which is an extension to sweep algorithm and are called petal algorithms. Here petals are analogous to feasible set of routes.

Petals are generated and final selection is made by solving a set partitioning model.

Foster and Ryan (1976) and Ryan et al. (1993) have developed heuristic rules for determining the set of routes to be selected, while Renaud et al. (1996a, 1996b) have described an extension consisting of two embedded or intersecting routes called two-petals. They are better in performance than sweep algorithm.

5.2.1.2.4 Bramel and Simchi-Levi (1995)

They discussed another two-phase method with a fixed number of vehicles. This algorithm determines route seeds by solving a capacitated location problem, where m customers are selected by minimising the total distance between each customer and its closest seed, and by imposing that the total demand associated with each seed be at most Q. Once seeds have been determined and the single-customer routes are initialised, the remaining customers are inserted in the current routes by minimising insertion costs.

Route-first, cluster-second algorithms

In these types of algorithm, initial route that visits all the nodes is constructed initially and later the route is subdivided into several routes starting from the depot. Haimovich and Rinnooy Kan (1985), and Bertsimas and Simchi-Levi (1996) proposed some of the research in this area.

5.2.1.3 Insertion heuristics

The main theme here is to start with an initial route and insert a node one by one. Solomon (1987) has proposed three types of insertion heuristics.

5.2.2 Improvement heuristics

These operate on each vehicle route taken on several routes at a time. Local search algorithms are often used to improve initial solutions generated by other heuristics. Starting from a given solution, a local search method applies simple modifications, such as arc exchanges or customer movements, to obtain neighbour solutions of possibly better cost. If an improving solution is found, it then becomes the current solution and the process iterates; otherwise a local minimum has been identified.

Among simple local search algorithms, the most famous are 2-Opt and 3-Opt. The 2-Opt algorithm was first proposed by Croes (1958), although the basic move had already been suggested by Flood (1956). In this algorithm, two edges are removed from an existing tour and two new edges connected. In 3-Opt (Bock, 1958; Lin, 1965), the exchange replaces up to three edges of the current tour.

Simpler local optimisation algorithms than 2- and 3-Opt exist. Bentley (1992) introduced 2.5-Opt algorithm in which we expand the 2-Opt neighbourhood to include 3-Opt move. In a 2.5-Opt move one relocates a single city from its current location to a position between two current tour neighbours elsewhere in the tour. k-opt is also similar to 2-Opt and 3-Opt with different value of k.

Lin-Kernighan: This is variable k-opt algorithm, it decides at every iteration the most suitable value.

5.2.3 Metaheuristics

5.2.3.1 Ant colony algorithms

Ant colony optimisation is a meta-heuristic technique that implements the nature of ants to solve optimisation problems. Whenever an ant moves, it releases pheromone, which when comes in contact with other ants who follow the pheromone, the decision has to be

made whether to follow it or not. If it follows the trail, the ant's own pheromone adds up to the existing trail, and thus the probability of the next ant choosing the same path increases. Pheromone is the main source of communication between ants. Therefore, the more ants that travel on a path, the more attractive the path becomes for next ants. Also an ant who randomly chooses a shorter route, marks its path twice, before other ants return. This directly influences the selection probability for the next ant leaving the nest. Over time, as more ants are able to complete the shorter route, pheromone accumulates faster on shorter paths and longer paths are less reinforced. The evaporation of pheromone also makes less desirable routes more difficult to detect and further decreases their use.

Bell and McMullen (2004) described a model to explain this. To select the next customer j, the ant uses the following probabilistic formula

$$j = \arg\max\left\{(\tau_{iu})(\eta_{iu})^{\beta}\right\} \text{ for } u \notin M_k, \text{ if } q \le q_0$$

where τ_{iu} is equal to the amount of pheromone on the path between the current location i and possible locations u. The value η_{iu} is defined as the inverse of the distance between the two customer locations and the parameter β denotes importance of distance. Locations already visited by an ant are stored in the ants working memory M_k and are not considered for selection. The value q is a random uniform variable [0, 1] and the value q_0 is a parameter. When each selection decision is made, the ant selects the arc with the highest value from j unless q is greater than q_0.

In this case, the ant selects a random variable to be the next customer to visit based on the probability distribution of p_{ij}, which favours short paths with high levels of pheromone:

$$p_{ij} = \frac{(\tau_{ij})(\eta_{ij})^{\beta}}{\sum_{u \notin M_k}(\tau_{ij})(\eta_{ij})^{\beta}} \text{ if } j \notin M_k \text{ otherwise } 0$$

Each ant has to make a decision either to follow the most optimum and recognised path or select a random path to follow based on a probability function. This selection process continues until each customer is visited and the tour is complete.

Local and global trail updating is done to improve future solutions.

Local updating is done using the following equation:

$$\tau_{ij} = (1-\alpha)\tau_{ij} + \alpha\tau_0$$

where α is a parameter responsible for speed of evaporation and t0 is an initial pheromone value.

For global trail updating is done using the equation:

$$\tau_{ij} = (1-\alpha)\tau_{ij} - \alpha(L)^{-1}$$

where L is the total distance computed as the objective function value for the complete route of the artificial ant.

5.2.3.2 Genetic algorithms

In genetic algorithm, a solution is analogous to a chromosome. It starts by generating multiple solutions known as 'chromosomes' which is called a population. Holland and Goldberg (1975) first introduced this technique.

The first process is the selection of individuals from the population based on fitness function. The second process is the recombination in which two distinct solutions are decomposed and then their parts are randomly mixed to create new individuals. To create one new individual mostly two parents are selected and their most desirable features are combined in order to create an offspring solution, i.e., the new individual. To increase the chance of high-valued offspring in this recombination process, more offspring are generated than the number of individuals in the parent population.

After the new population is generated, the third process i.e. the mutation starts for each individual. Due to this, the solution becomes different and thus escapes local minima. Final modified population results after these processes. Now again the selection of a new population starts while saving the current best solution. Every offspring individual is rated by one or more criteria and the best or most promising individuals are promoted to a new parent population.

Algorithm steps:

1 Generate an initial population of individuals.

2 Evaluate the fitness by making use of function of all individuals.

3 Select fit individuals for reproduction.

4 Recombine between individuals.

5 Mutate individuals.

6 Evaluate the fitness using fitness function of the modified individuals.

7 Generate a new population.

8 If termination conditions meet, then stop otherwise go to Step 3.

5.2.3.3 Simulated annealing

Simulated annealing is a metaheuristic based on ILS procedure that tries to avoid local optima by accepting probabilistically moves to worse solutions. Annealing represents a physical process in which temperature of a solid is increased until it melts in a heat bath, then allowed to cool slowly until it solidifies into a low-energy state. The method was independently described by Kirkpatrick and Vecchi (1983) and Černý (1985).

According to this, a solution is analogous to the state of a solid and the cost function to its energy. At each iteration, the existing solution is improved. A solution is always modified to a better one, but also a modification could be accepted even if the solution cost increases. A probability criterion is defined in which if the 'temperature' parameter is high and the cost increase is low, then it is modification is accepted. When the temperature is sufficiently low, only improving modifications can be accepted and the method stops in a local minimum.

Algorithm steps:

1 Select starting temperature and initial parameter values.

2 Randomly select a new point in the neighbourhood of the original.

3 Compare the two points using the fitness function.

4 Repeat Steps 2 and 3 until system reaches equilibrium state.

5 Decrease temperature and repeat the above steps, stop when system reaches frozen state.

5.3.2.4 Tabu search

Tabu search is a memory-based search strategy to guide the local search method to continue its search beyond a local optimum. Glover (1986) described it as "tabu search is based on introducing flexible memory structures in conjunction with strategic restrictions and aspiration levels as a means for exploiting search spaces".

It starts with an initial solution to the problem i.e. current solution, and searches for the best solution in a suitably neighbourhood of the solution. Then the best solution becomes the current solution and the process repeats itself. When certain conditions have been met like execution time conditions or solution quality objectives, the algorithm terminates. Tabu search maintains a list of neighbour generation moves that it considers forbidden, or tabu in order to prevent it from considering earlier solutions. Tabu tenure of the move is the number of iterations till which it has to stay in the tabu list. Therefore during the execution of the search, the list changes continuously making the search an adaptive memory search algorithm.

1 Choose an initial solution.

2 Create a list of solutions which are not in the tabu list and evaluate them.

3 Choose the best admissible solution.

4 If a terminating condition is met then stop, final solution is obtained.

5 Else update tabu and aspiration conditions and repeat Step 2.

5.3.2.5 Memetic algorithms

Memetic algorithms are used as a combination of evolutionary approaches with separate individual learning or local improvement procedures for problem search. Basically memetic algorithms are extension of genetic algorithm by applying a local search on individuals after mutation. Following biological evolution, the transition took place from genes to memes.

Memetic algorithms by Dawkins (1976) denote a family of meta-heuristics that have as central theme the hybridisation of different algorithmic approaches for a given problem. In Dawkins' words:

> "Examples of memes are tunes, ideas, catch-phrases, clothes fashions, and ways of making pots or of building arches. Just as genes propagate themselves in the gene pool by leaping from body to body via sperms or eggs, so memes propagate themselves in the meme pool by leaping from brain to brain via a process which, in the broad sense, can be called imitation."

A meme is a perceptive arrangement that can be transmitted from one individual to another one. Replication is done by transmitting the meme to the memory of another individual i.e. carrier without erasing the meme previous carrier individual. MAs exploit problem and instance-based knowledge by incorporating pre-existing heuristics, pre-processing data reduction rules, approximation, local search techniques, specialised recombination operators etc. This exploitation is an important characteristic of memetic algorithms.

Runggeratigul (2004) described the memetic algorithm as:

1 Initialisation: generate feasible individuals for the initial population.

2 Local search: apply local search.

3 Parent selection: select pairs of parents for the crossover process.

4 Crossover: produce two children from each pair of parents.

5 Mutation: perform mutation on each child (repair infeasible child if needed).

6 Local search: apply local search to each child after mutation.

7 Population replacement: select children for the population in the next generation.

8 Termination: If the predefined number of generations have been completed.

9 STOP. Otherwise, go to parent selection step.

5.3.2.6 Variable neighbourhood search

VNS is a recent metaheuristic which exploits systematically the idea of neighbourhood change, both in the descent to local minima and in the escape from the valleys which contain them (Hansen and Mladenović, 1997).

The algorithm uses local search to get from these neighbours to local optima. It explores increasingly distant neighbours of the incumbent solution so, it does not follow a single trajectory like simulated annealing and tabu search algorithm, but changes its course systematically to other neighbourhood and updates from this solution to a new one if and only if there is an improvement in the solution. To change the neighbourhood firstly, it descents to find a local optimum and then takes a jump to get out.

Steps:

1 Initialisation: select a set of neighbourhood structures and find an initial solution. Also choose a stopping condition.

2 Generate a point at random.

3 Apply a local search method starting with the generated point as initial solution to find local optimum.

4 If local optimum is better than the incumbent, then move there and continue the search until the terminating conditions achieve.

5 Otherwise go back to Step 2.

The important extension of this algorithm is variable neighbourhood descent algorithm in which the exploration and change of neighbourhood is done simultaneously with the local search procedures.

6 Conclusions and future scope of research

A total of 117 research articles were screened, 37 are from CVRP, 19 are from MDVRP and 61 from VRP with pickup and delivery. Further dissecting the findings, in CVRP, seven papers have implemented exact algorithms, 20 have used various heuristics and ten studies consist of hybrid use of several algorithms including exact and heuristics.

In MDVRP, only one article has used exact algorithm, 12 applied heuristics and six hybridised various algorithms. Similarly, in VRP with pickup and delivery, eight papers using exact, 38 using heuristic and 16 articles making use of hybrid algorithms have been presented in the last few decades.

Table 6 Findings

Problem		*Exact*	*Heuristics*	*Hybrid use*	*Total*
CVRP		7	20	10	37
MDVRP		1	12	6	19
VRPPD	VRPSPD	5	13	13	31
	VRPB	2	12	2	16
	VRPMPD	0	13	1	14
Total		15	70	32	117

It is evident from the review that most of the work has been presented in the field of heuristics but as the trends suggest, researchers are attempting to make use of hybrid exact and heuristics algorithms to solve optimisation problem.

In the last few decades, there have been recent developments in the field of VRP due to improved mathematical programming tools and practical importance in the field of logistical management. In this review, we kept our focus on three variants of VRP namely CVRP, MDVRP and VRP with pickup and delivery. From the review, it is clear that few researches have been carried under hybrid methods in comparison to heuristics algorithms. Hence, future researchers may focus on using an amalgamation of exact and heuristics algorithms so that good as well as best possible optimum solution can be determined.

A limitation of this study is the size of the survey. It covers only a limited set of journals. It is quite likely that some research concerning our study is published in journals that were not included in our review. So, the scale of the review should be extended in order to strengthen it. We also did not include detailed classification of these algorithms so one can also do some work relating to this.

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